Team Name : TEAM LOGIC LORDS.

Team Leader Name : KHUSHBOO MAHATO

Problem Statement : NOVEL METHOD TO DETECT LANDSLIDES ON EARTH & MOON :USING INDIAN REMOTE SENSING IMAGES

# Team Members

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# 💡 Brief about the Idea:

Our project idea is basically emphasized on improvising the existing ideas on novel method to detect landslides on earth and moon using Indian remote sensing images.

The goal is to automate the detection process using a combination of image processing and AI-models . We aim at providing a hybrid ML-GIS Framework that uses multi temporal spectral indices and terrain change analysis.

The satellite CARTOSAT having sensors LISS –III & LISS- IV have already been implemented on this problem consist of the OHRC camera of chandrayaan -2 that was crucial for obtaining high resolution images .

Our motive emphasises on increasing / improvising the quality of camera such as PRISM , ENMAP that will provide deep spectral information .

# Opportunity should be able to explain the following:

- **How different is it from any of the other existing ideas ?**

  1. Landslides detection on the moon
  2. Use of Indian remote sensing data.
  3. Dual planet comparative approach.

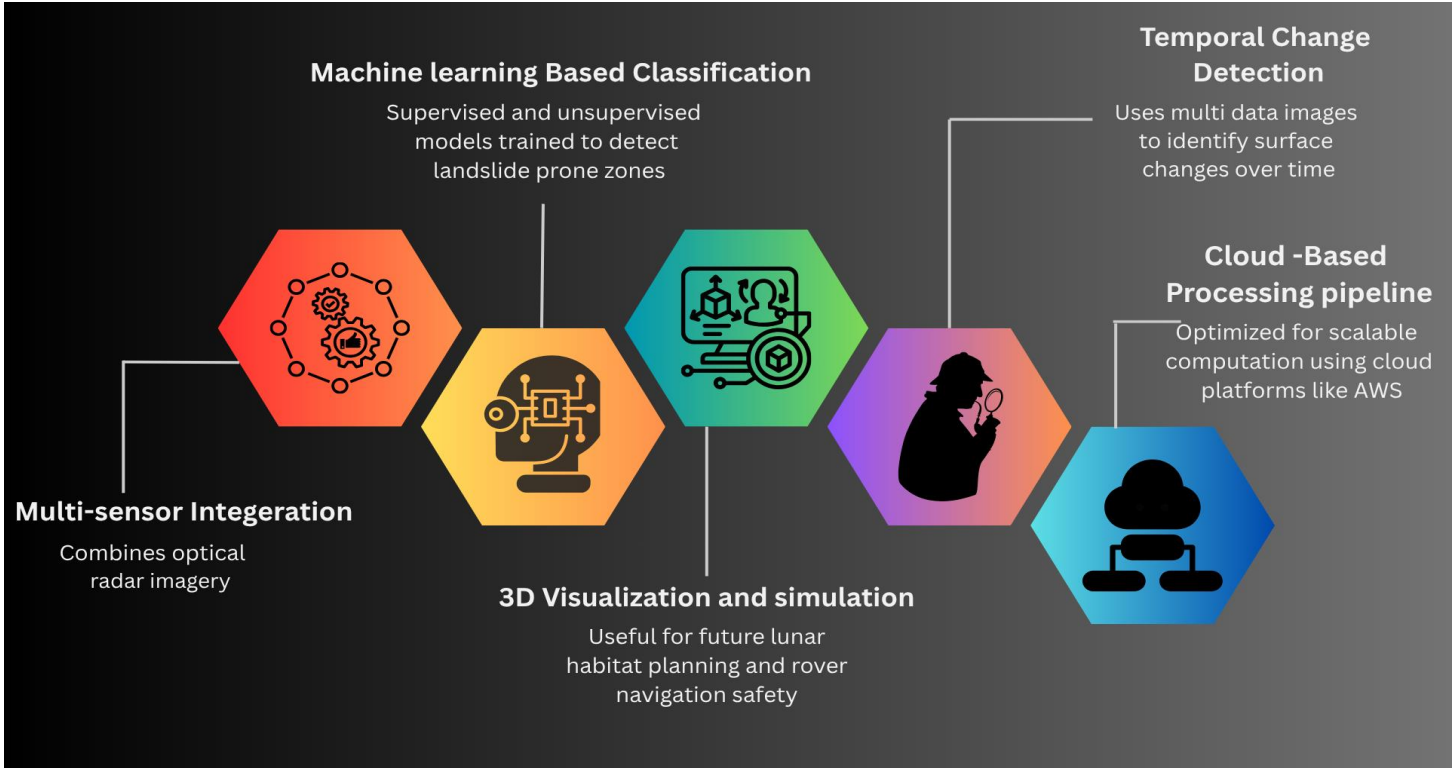
- **How will it be able to solve the problems ?**

  1. Integration with HYSIS data to limit the respective spectral bands compared to hyperspectral sensors.
  2. Application of topographic correction algorithm to enhance preprocessing .
  3. Employ unsupervised in learning techniques for anomaly detection in lunar terrain.

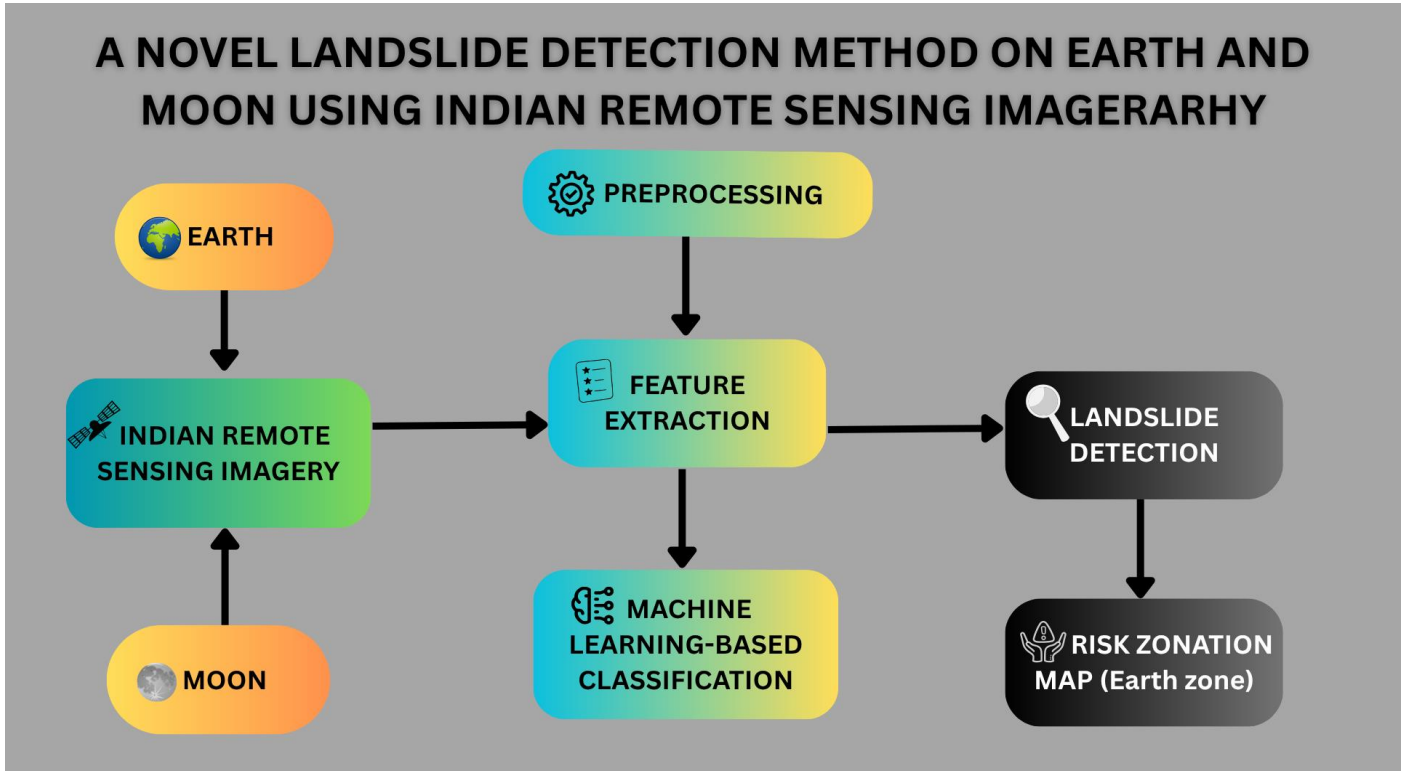
- **USP of the proposed solution ?**

  1. Enhanced dual- channel mode for landslide detection framework .
  2. Utilization of Indian remote sensing imagery sensing for detection of earth from space and laser scanning and ground-based interferometry.

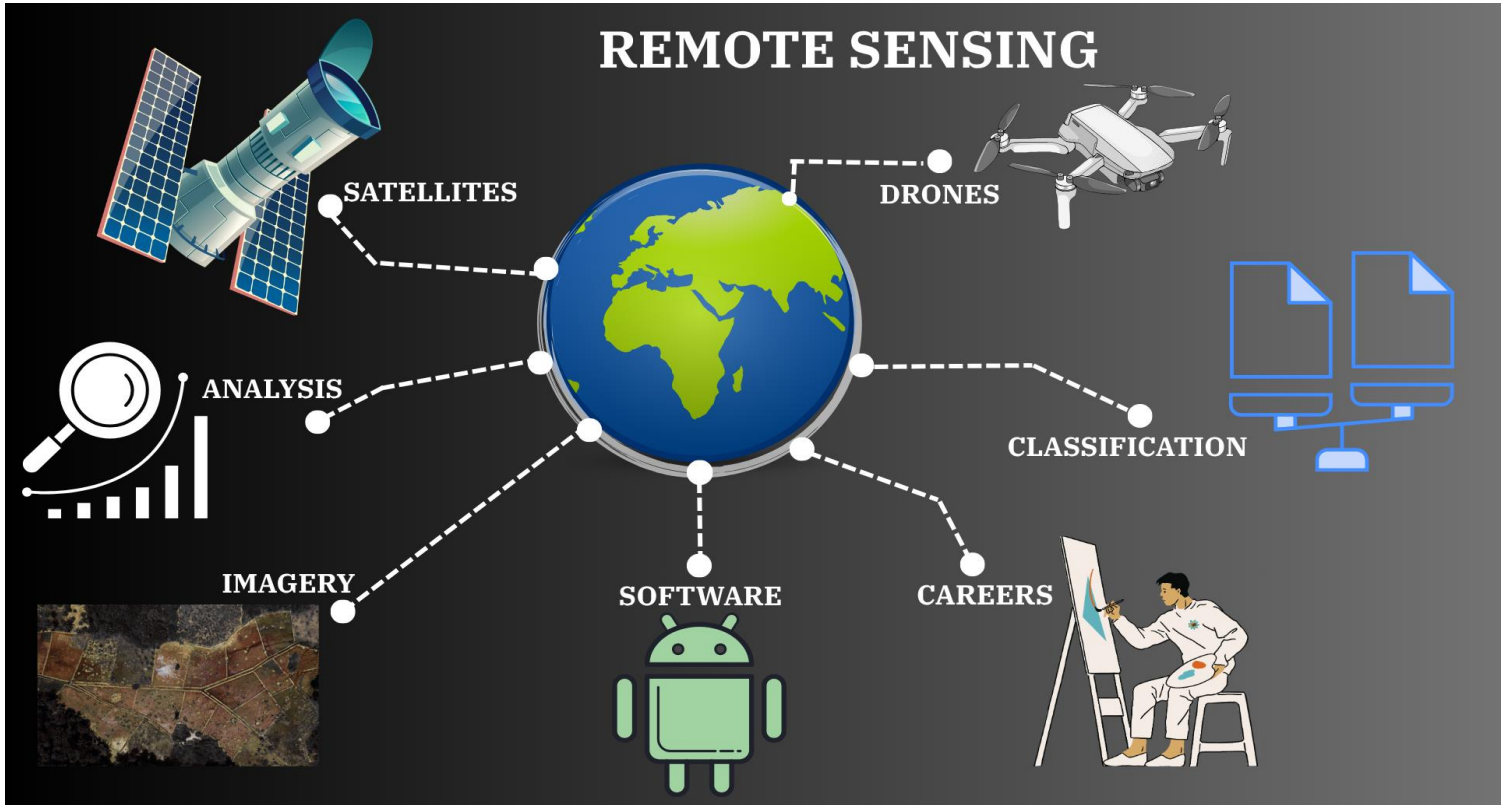
# List of features offered by the solution.



# Process flow diagram or Use-case diagram



# Architecture diagram of the proposed solution



# Technologies to be used in the solution:

| | |
|---|---|
| TECHNOLOGY USED | : PYTHON , SNAP ( Pre processing) |
| LIBRARY | : NUMPY , RASTERIO |
| MACHINE LEARNING | : TENSOR FLOW , PYTORCH |
| DETECTION & IMAGE ANALYSIS | : OPEN CV , RASTERIO |
| GIS & MAPPING | : LEAFLET.js , QGIS |
| FRONTENT | : HTML, CSS , JAVASCRIPT, REACT.js |
| DATABASE | : SQL |
| CLOUD | : AWS |

## Potential Application

- Early warning systems for Himalayan and India slopes
- Lunar geohazard mapping for ISRO's future human mission
- Climate –linked landslide modelling (Earth )
- Identifying unstable craters or rilles(Moon)

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# THANK YOU